Register Variation in Lexical Diversity and Part-of-Speech Distribution: A Comparison of Scientific Abstracts and Newspaper Articles

Regina Kukuh Prasetyo Ati

ABSTRACT

This research investigates register variation in lexical diversity and part-of-speech distribution in scientific abstracts and newspaper articles. To understand these differences, the author compares scientific abstracts and newspaper articles to identify differences in word usage and word types between the two text types. The research method involves collecting and analyzing data, consisting of abstracts from scientific articles published in scientific journals and newspapers. The analysis reveals significant differences: scientific abstracts exhibit lower lexical diversity and tend to use specific word types more frequently than newspaper articles. The author explores the implications of these findings in the context of scientific and journalistic communication. The author concludes that differences in register and language use between scientific abstracts and newspaper articles reflect different communication needs in the two contexts. This research provides additional insights into the linguistic characteristics of both text types and highlights the importance of understanding appropriate language use in different communication contexts.

Keywords: Register Variation, Lexical Diversity, Part-of-Speech Distribution, Scientific Abstract, Newspaper Article

INTRODUCTION

Language is a highly complex communication tool, with diverse variations and styles depending on the context and communicative situation. One important aspect of language variation is register, which refers to the use of language appropriate to specific social, cultural, and situational contexts (Biber & Egbert, 2023). In this context, register is often considered one of the elements determining a language's level of formality or informality. In linguistic studies, register variation becomes crucial for understanding how humans interact and convey messages in various communication contexts (Gemmell, 2009).

According to Ogemdi et al. (2021), register variation in language encompasses the use of vocabulary, sentence structures, intonation, and different language styles depending on the situation and intended audience. Formal settings (speeches, academic writing) elicit formal vocabulary, complex structures, and elevated styles. Conversely, informal settings (conversations) encourage relaxed language, everyday vocabulary, and simpler structures.

The importance of understanding register variation in language is limited to one's ability to communicate effectively in various contexts and can influence perceptions and relationships among individuals in society (Moreno, 2006). A good understanding of how and when to use appropriate registers can help individuals avoid communication errors or misunderstandings that may arise from inappropriate language use.

Research on register variation in language holds significant importance in understanding the differences in language usage among specific genres. Understanding the differences in language usage among specific genres helps us appreciate the complexity of language and culture around us. Additionally, research on register variation can also provide insights into the social, cultural, and ideological identities embedded in the language use of individuals or specific groups (Agha, 2004).

Lexical diversity plays a crucial role in representing register variability in language. In this context, lexical diversity refers to the various choices of words, phrases, and vocabulary available to speakers to express themselves (Baese-Berk et al., 2021). With lexical diversity, speakers can choose words that best suit their communicative purposes and adjust their language style according to their communication context. Lexical diversity enables speakers to adapt their language register to their communication needs flexibly, which, in turn, aids in achieving effective communicative goals (Ravid & Tolchinsky, 2002). Furthermore, part-of-speech distribution is crucial in understanding the language used in various communication contexts. The part-of-speech distribution reflects the tendency to use specific words in a text, which can provide valuable clues about the characteristics of the observed register (Rovenchak & Buk, 2018). According to Munro (2006), by analysing part-of-speech distribution, we can identify typical patterns of nouns, verbs, adjectives, adverbs, and other types of words in a text. This provides a deeper understanding of the language structure used in various communication contexts, allowing us to distinguish between formal and informal language or different registers within the same communication domain.

The need for research comparing scientific abstracts and newspaper article registers is becoming increasingly important, considering the significant differences in language usage and its effects on readers or listeners. Research comparing these two types of registers can provide insights into differences in language usage that can affect communication effectiveness and understanding in academic and public contexts. With a better understanding of these differences, we can identify more effective communication strategies for conveying information to various audiences, enhancing understanding and community participation in various science and daily life fields.

The analysis will focus on scientific abstracts and newspaper articles to ensure text length equivalence. While scientific articles provide valuable data, their longer texts compared to newspaper articles can introduce bias in the analysis. Analyzing abstracts, which summarize core information and findings, allows for a more controlled and fair comparison. This research will be based on the register theory, which states that language usage varies based on the communicative context (register) in which it occurs.

The register theory pioneered by Halliday (1978) serves as the main foundation of this research. This theory suggests that language usage varies depending on the situation and context surrounding communication and emphasizes that language usage evolves according to its situation and context. This research demonstrates how vocabulary and sentence structures adapt to different communication situations by comparing scientific and journalistic registers.

The main concepts in Halliday's register theory (1978) are as follows: Firstly, register is understood as functional variations of language. This means how we use language will change depending on its context. Secondly, situational context is a crucial element influencing language usage. Thirdly, register analysis is conducted by analyzing these three elements (field, tenor, and mode), thus enabling us to understand how language choices shape the meaning and function of a text.

Halliday's register theory is valuable for understanding how language functions in various contexts. This theory is applied in various fields, including applied linguistics, writing instruction, and discourse analysis. By understanding this theory, we can gain insights into the varied use of language in different communication situations.

Exploring these differences holds significance across various disciplines. In cognitive science, it can elucidate the reading process for diverse text types. In Natural Language Processing (NLP), it can inform the development of automated text classification tools. Within language pedagogy, such knowledge can guide instructors in tailoring instruction to equip students with vocabulary and grammar specific to different contexts. Finally, this research strengthens language theory by offering empirical support for register variation. It demonstrates how specific word choices and their grammar categories contribute to the different characteristics of scientific discourse and newspapers.

Several studies have explored how language use changes across different contexts, known as registers. Biber and Conrad (2005) propose a framework for studying this variation using corpus linguistics, analyzing features like vocabulary and grammar, focusing on factors like setting and purpose. Gray (2013) argues that discipline is not the only factor influencing academic writing, highlighting the role of research methods and paradigms. Kerz and Wiechmann (2015) examine how specific grammatical constructions vary between academic and newspaper writing. Kettunen (2014) explores the potential of a single measure, the type-token ratio, to reflect language complexity across registers. Finally, Weizman (1984) and Youmans (1990) investigate journalistic language, with Weizman searching for universal features and Youmans proposing a method to analyze lexical style.

This research directly contributes to the understanding of register variation. This research focuses on two distinct registers with specific communicative goals by comparing scientific abstracts and newspaper articles. Analyzing lexical diversity (vocabulary richness) and part-of-speech distribution (using different word types) will reveal how language adapts to convey complex scientific information versus current events clearly and concisely. This contributes to our knowledge of how register influences vocabulary choice and the fundamental building blocks of sentences. This research aims to provide empirical evidence supporting register theory by demonstrating how specific word choices and grammar categories contribute to different characteristics of scientific discourse and newspapers. This research is expected to contribute to understanding how language adapts and functions in different communicative contexts. Furthermore, this research can benefit researchers in tailoring their writing for specific audiences and improving the communication of complex scientific findings to the public.

METHOD

This research employs a multi-stage methodology to compare the language characteristics of scientific abstracts and newspaper articles. The first stage involves data acquisition, where a corpus of 150 scientific abstracts and 150 newspaper articles will be obtained from Kaggle.

Next, data processing and analysis will be conducted using SpaCy, a Python library. Text cleaning and tokenization will be performed, including stop word removal. Lexical diversity will then be assessed by calculating each text category's Type-Token Ratio (TTR). Additionally, spaCy will be used for part-of-speech (POS) tagging to determine the grammatical function of each word. The frequency distribution of POS tags will be analyzed for both scientific abstracts and newspaper articles.

A comparative analysis will be conducted to evaluate potential differences in vocabulary richness (TTR) and grammatical structure (POS distribution) between the two text categories. The findings will be analyzed to investigate their alignment with Halliday's register theory, which posits that language use adapts to the communicative situation (register). Finally, data visualization will be employed using Matplotlib to enhance the clarity of the analysis, aiming to provide a deeper understanding of the distinct language characteristics employed in scientific abstracts and newspaper articles.

ANALYSIS

The analysis conducted in this research closely relates to register theory by Halliday (1978), which identifies three key factors influencing language choice. The first is field, referring to the subject matter (scientific versus journalistic). Scientific abstracts tend to have a narrower field, using specialized technical vocabulary specific to the scientific domain. On the other hand, newspaper articles cover a wider range of topics, employing a more general vocabulary for a wider audience, leading to higher lexical diversity.

The second factor is tenor, which pertains to the relationship between participants (formal in scientific discourse and possibly more informal in news discourse). Scientific abstracts are usually written in a formal tenor, with the use of non-subjective language and less use of adverbs. Newspaper articles can vary in tenor depending on the specific type of article (e.g., editorial versus sports report).

The third factor is mode, which is how communication occurs (written in both cases). Scientific abstracts and newspaper articles are both written texts (mode), but the written mode itself can influence language use. For example, both may employ more complex sentence structures than spoken language.

The analysis of differences in lexical diversity and part-of-speech distribution between scientific articles and newspapers can be connected to and support Halliday's register theory in several ways. Firstly, lexical diversity, field, and tenor indicate significant differences. Scientific abstracts tend to have a lower Type-Token Ratio (TTR), indicating the use of a more limited vocabulary. Conversely, newspaper articles, aimed at a general audience, tend to have a higher TTR due to the need for a broader language variety.

Secondly, different patterns are observed in part-of-speech distribution and mode (channel). Scientific abstracts tend to have higher numbers of nouns and adjectives, emphasizing entity descriptions and their attributes. This aligns with the written mode of scientific communication. On the other hand, newspapers tend to have higher numbers of nouns and verbs, focusing on events and actions, and reflecting a more engaging and broader narrative style.

These findings support the idea in Halliday's register theory that language choices are influenced by the situation. Scientific abstracts tend to use concise technical language, while newspapers employ a narrative style with a broader vocabulary. The differences in language use reflect the core principle of Halliday's register theory that language adapts to the context.

Furthermore, differences in register variables indicate how scientific abstracts and newspaper articles, representing different registers, exhibit different language patterns regarding vocabulary and grammar. This supports the idea that language adapts to context and communication functions.

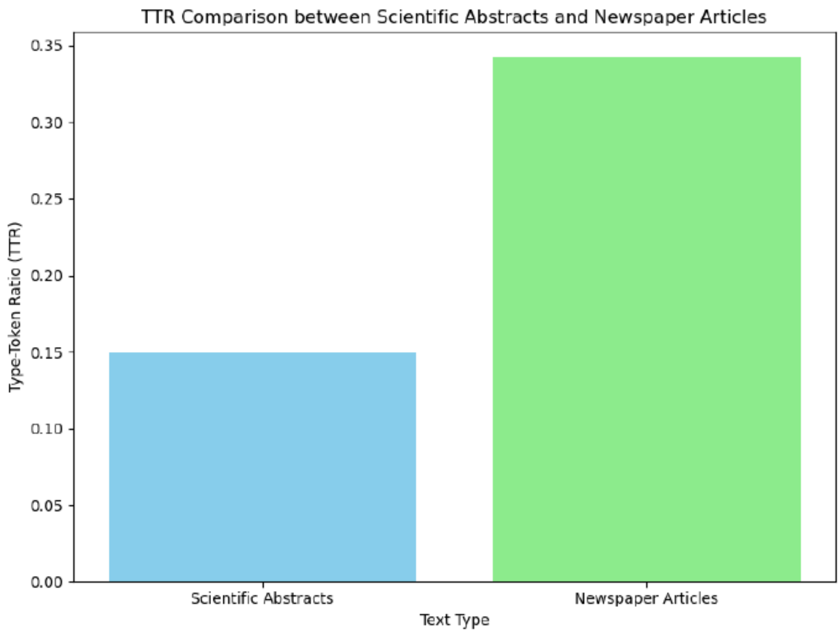
Context, participants, and communication purposes also play significant roles in language choice. Scientific abstracts, written in a formal academic context by scientists for a specific audience, aim to convey factual information accurately. Newspaper articles, written for a general audience in a less formal setting, aim to provide information, persuade, or entertain.

This analysis confirms Halliday's register theory that language choice is driven by context and communication function. It emphasizes that language adapts to different situations, roles, and communication purposes.

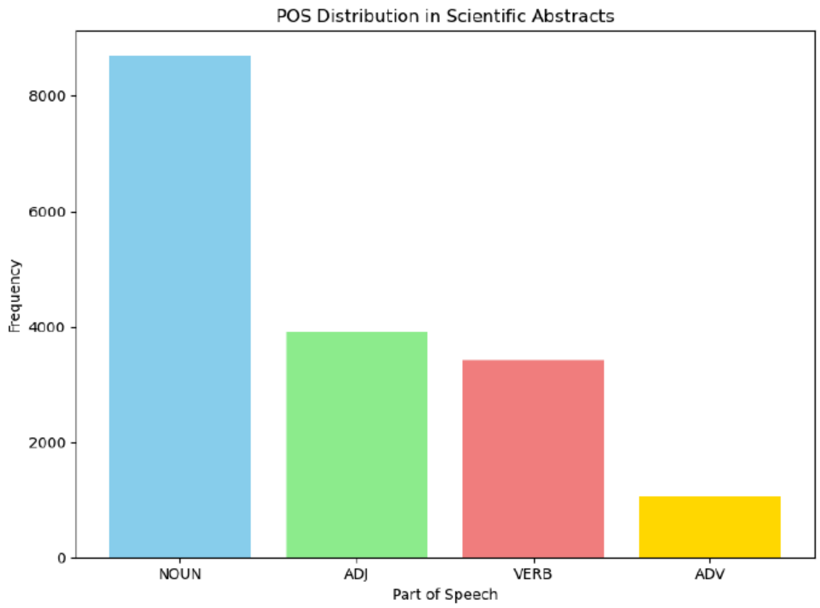
RESULTS

The analysis revealed significant differences in register between scientific abstracts and newspaper articles, supporting Halliday's register theory. A dataset of 150 scientific abstracts and 150 newspaper articles was obtained from Kaggle. This data was then processed using SpaCy, a Python library for natural language processing, to calculate the Lexical diversity as measured by the Type-Token Ratio (TTR), and part-of-speech distribution between scientific abstracts and newspaper articles.

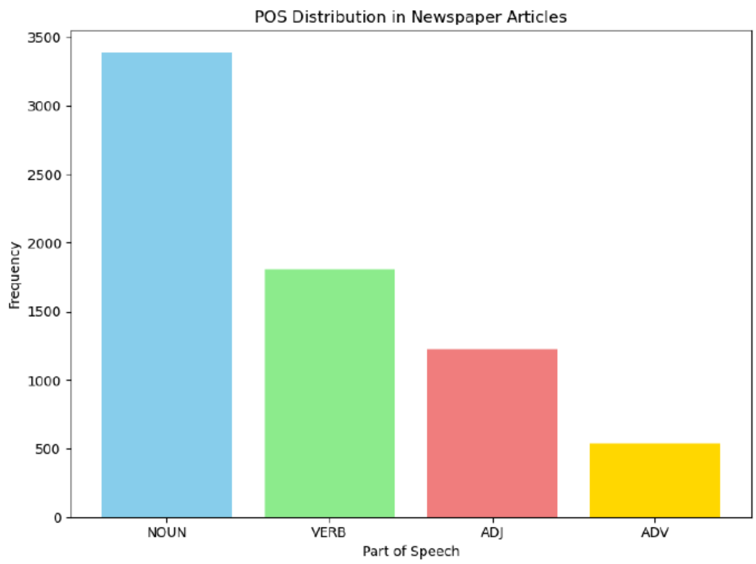
Scientific abstracts exhibited lower lexical diversity, signifying a reliance on a more technical vocabulary with less variation in word choice. This suggests a more restricted set of terms specific to the scientific field. Conversely, newspaper articles displayed higher lexical diversity, indicative of a broader language variety. This reflects the need to cater to a general audience with a wider range of vocabulary and background knowledge. The type-token ratio (TTR) in the newspaper articles is 0.3417742888494978, whereas in scientific abstracts, the TTR is 0.1492271105826397. These findings depict that scientific abstracts tend to have lower lexical diversity than newspaper articles, as visualized in the following chart.



In part-of-speech distribution, focusing on nouns, adjectives, verbs, and adverbs, the analysis revealed that scientific abstracts tend to have higher numbers of nouns and adjectives, emphasizing entities and describing characteristics relevant to the scientific field. On the other hand, newspapers tend to have higher numbers of nouns and verbs, focusing on events and actions, and reflecting a more engaging and broader narrative style clearly and concisely for a wider audience. The following column charts depict the findings, focusing on the distribution of nouns, adjectives, verbs, and adverbs within the PoS analysis.



In scientific abstracts, nouns (NOUN) are the most dominant part of speech with a frequency of 8691 occurrences. This suggests a primary focus in scientific abstracts on concepts, objects, and phenomena under study. Furthermore, adjectives (ADJ) occupy the second position with a frequency of 3913 occurrences, indicating a tendency to provide descriptions or characteristics of the objects or phenomena being researched. Verbs (VERB) show a frequency of 3426 times, indicating actions or processes as the main focus in expressing research findings. Adverbs (ADV) have frequencies of 1068, adding additional information in sentences.



Meanwhile, the results of the analysis of part-of-speech distribution in newspaper articles show a different pattern. The collected data shows that nouns (NOUN) are the most dominant part of speech with a frequency of 3381 times, followed by verbs (VERB) with 1810 times. The number of adjectives (ADJ) is also quite significant, with a frequency of 1406 and adverbs with a frequency of 538. From these results, it is apparent that newspaper articles use a wide variety of part-of-speech distribution, with a strong focus on events and actions that are marked by the use of verbs, reflecting a more engaging and broader narrative style, while scientific abstracts tend to have a higher focus on the use of specific and technical vocabulary. These findings indicate differences in communication strategies and writing styles between the two text types.

DISCUSSION

The analysis results indicate significant differences in lexical diversity and part-of-speech distribution between scientific abstracts and newspaper articles. This discussion compares registration variations in lexical diversity and part-of-speech distribution between scientific abstracts and newspaper articles. The data shows significant differences in the pattern of word type usage between the two genres.

The type-token ratio (TTR) research findings reveal significant differences in lexical diversity between scientific abstracts and newspaper articles. The TTR for newspaper articles is 0.3417742888494978, whereas the TTR for scientific abstracts is only 0.1492271105826397. This difference indicates that scientific abstracts tend to have lower lexical diversity than newspaper articles.

In the part-of-speech distribution, scientific abstracts show a significant tendency to use nouns, followed by adjectives, verbs, and adverbs, with frequencies of 8691, 3913, 3426, and 1068, respectively. These results are consistent with previous findings stating that scientific abstracts tend to use more nouns to convey technical and conceptual information (Biber & Conrad, 2005). Additionally, using adjectives and verbs in scientific abstracts can also be explained as an effort to provide accurate descriptions of the reported research (Gray, 2013).

On the other hand, newspaper articles exhibit a different pattern in the part-of-speech distribution. Nouns still dominate, albeit with lower frequencies than in scientific abstracts, with 3381 occurrences. However, verbs take the second position in newspaper articles, with 1810 occurrences indicating a tendency to emphasize actions or events in journalistic narratives (Weizman, 1984). The usage of adjectives and adverbs is also more limited in newspaper articles, with 1222 and 538 occurrences, respectively. This aligns with previous research stating that newspaper articles tend to focus more on facts and information directly related to the events being discussed (Kerz & Wiechmann, 2015).

This comparison highlights significant differences in the pattern of word type usage between scientific abstracts and newspaper articles. While scientific abstracts tend to be more technical and descriptive by using more nouns and adjectives, newspaper articles emphasize actions and facts directly related to the events being reported by using more nouns and verbs. This understanding is essential for recognizing the characteristic language of each genre and can serve as a guide for writers and translators in producing texts suitable for the desired context (Baker, 2000).

These findings are consistent with previous research by Kettunen (2014), who also found that more formal text types tend to have lower TTRs than more informal text types. Additionally, research by Youmans (1990) suggests that scientific articles generally tend to use more specific and limited vocabulary, which may explain why the scientific abstracts in this research have a lower TTR compared to newspaper articles.

The differences in lexical diversity between these two types of texts could be influenced by their respective communicative purposes. Newspaper articles are usually written for a broader audience and aim to provide information quickly and clearly. Conversely, scientific abstracts are intended for readers with a scientific background and may require specialized terminology not commonly used in everyday language. This aligns with research findings by Cambre et al. (2020), which indicate that texts aimed at a narrower audience tend to have lower TTRs due to more technical terminology.

In addition to lexical diversity, part-of-speech distribution is also a concern in this analysis. Although there are no significant differences in part-of-speech distribution between the two types of texts, small differences can be observed. Newspaper articles tend to have a higher proportion of nouns and verbs, while scientific abstracts have a slightly higher proportion of nouns and adjectives. This is consistent with the notion that scientific abstracts tend to emphasize conceptualization and description of observed phenomena, thus tending to use nouns and adjectives to express related concepts and information. On the other hand, newspaper articles, which typically aim to provide a more dynamic and familiar overview of current events, tend to use nouns and verbs to present information more dynamically and actively. Although these differences are not statistically significant, they may reflect the communicative purposes of each type of text.

However, this research has several limitations, one of which is the limited sample size, especially regarding the number of newspaper articles and scientific abstracts analyzed.

Additionally, this research only considers lexical diversity and part-of-speech distribution as single indicators of differences between the two types of texts. Further research could consider other factors such as sentence length, information density, and overall text structure to better understand the differences in language use between newspaper articles and scientific abstracts.

CONCLUSION

In this research, the author investigates register variation in lexical diversity and the part-of-speech distribution between scientific abstracts and newspaper articles. The author found significant differences between the two types of texts through the analysis. The research findings indicate that scientific abstracts tend to have lower lexical diversity than newspaper articles. This may be due to technical terms and specialized vocabulary, which are more commonly found in scientific contexts. The part-of-speech distribution also shows differences between scientific abstracts and newspaper articles. Scientific abstracts tend to use specific word types, such as nouns and adjectives more uniformly, while newspaper articles tend to use specific word types, such as nouns and verbs. These findings illustrate that language usage in scientific and journalistic contexts exhibits different patterns, with scientific abstracts showing a tendency to use more consistent and technical language. The implications of these findings underscore the importance of understanding the context and communication goals in assessing lexical diversity and the part-of-speech distribution in texts. Further research could explore the factors influencing these differences and their relevance in scientific communication and journalism. Thus, this research contributes significantly to understanding register variation in written language and its implications in various communicative contexts.

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